

K-Bound: When Is Label-Free Adaptation Knowable?

Helpful, Harmful, and Unknowable Regimes Under Distribution Shift

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Abstract

An unlabeled adaptation procedure can produce an update without establishing that deploying it will help. We study when the available evidence supports ADAPT, FREEZE, or ABSTAIN for a fixed model pair. For binary 0/1 loss and positive disagreement mass, K-Bound characterizes the exact compatible-benefit interval under a declared bound β on a hidden target calibration residual. In the full label-kernel class, a strict direction is uniformly justified exactly when $|M| > \beta$, where M is the observable disagreement margin. A complementary evidence-fibre theorem gives the smallest residual bound that any uniformly valid audit of the same information must cover. Additional processing cannot remove ambiguity left by that information. Knowability-Guided Adaptation (KGA) addresses a separate empirical target: the measured score difference between a candidate and its frozen baseline on an evaluation cell. Labeled development and disjoint residual-calibration cells fit its benefit predictor and error interval; the action uses no scored-cell outcome. On the opened, dependent six-family CIFAR-10-C diagnostic, Tent and EATA have lower point regret than both fixed policies. Tent makes 2 false ADAPT decisions among 1,091 ADAPT actions and 1 false FREEZE among 337 FREEZE actions; EATA makes neither error among 1,207 ADAPT and 105 FREEZE actions. SAR instead favors always-adapt. On CCT-20, all 45 cells retain the frozen model, including one false FREEZE; five-checkpoint DomainNet development evaluation yields universal abstention. Thus these natural-shift studies support conservative retention, not useful selective routing. The theory distinguishes structural ambiguity

from prediction uncertainty; the experiments do not establish coverage or population protection on unseen shifts. **Keywords:** test-time adaptation; adaptation gating; abstention; partial identification; distribution shift; conformal calibration; selective decisions.

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1 Introduction

Test-time adaptation changes a model using unlabeled deployment data. Entropy minimization, filtering, regularization, and related methods can recover accuracy under shift. The same update can also make predictions worse under unfavorable shifts (Wang et al., 2021; Niu et al., 2022, 2023; Wang et al., 2022). The central deployment question is therefore directional action correctness: an adapter supplies a candidate, not a reason to deploy it.

Our primary research question is: *When can evidence available without new target labels justify replacing a frozen predictor with a fixed adapted candidate?* Let f_0 be the frozen predictor and f_a the candidate. Their population benefit is

$$\Delta = R_T(f_0) - R_T(f_a). \quad (1)$$

A positive value favors the update; a negative value favors the frozen model. We use three decisions. ADAPT asserts positive benefit, FREEZE asserts negative benefit, and ABSTAIN asserts

neither. Both non-adapt decisions serve f_0 , but only FREEZE makes a directional claim.

K-Bound treats this as partial identification. In a binary model under 0/1 loss, we fix the input law, the predictors, and every label-free observable. We then vary the target label kernel within a declared class. The compatible benefits give an exact frontier for uniform strict decisions. The class is indexed by a bound on a hidden target calibration residual. That bound is an assumption, not a quantity supplied by an unlabeled batch. A second theorem shows that an audit using the same evidence must cover the residual range that the evidence leaves unresolved.

Knowability-Guided Adaptation (KGA) is the empirical companion. It predicts a model pair’s measured benefit on an evaluation cell. It acts only when a residual-calibrated interval excludes zero. KGA is label-free at the scored decision, not during development and calibration: labeled development cells train the predictor, and disjoint residual-calibration cells calibrate its errors. The specified cross-fit excludes each scored cell’s outcome from its own prediction, radius, and decision. This computational separation does not make an already opened retrospective panel prospective; outcome access is reported separately for each study. K-Bound is the population partial-identification framework; KGA is the empirical cell-benefit interval gate. They share three-action decision logic, but not a target or guarantee: K-Bound reasons about population benefit Δ through (M, γ, β) , whereas KGA predicts measured cell benefit Δ^{cell} through $(\hat{\Delta}, \varepsilon)$.

1.1 Three Scientific Contributions

1. **Exact population frontier.** We construct target label laws with identical label-free evidence, derive their feasible benefit interval, and characterize its strict positive, strict negative, and unresolved regions, including the zero-benefit boundary.
2. **Evidence-fibre audit limit.** We identify the least residual bound that the same observable law can support uniformly, explaining why historical labels help only through a stated link to the deployment domain.
3. **Empirical paired-benefit gate.** We give a cell-outcome-disjoint three-way cross-fit for the controlled grids and evaluate its ADAPT/FREEZE/ABSTAIN behavior on constructed shifts and its retention behavior on natural shifts. We report action exposure and false-action denominators before secondary accuracy and regret consequences. The standard coverage-to-action implication states exactly what follows if the interval covers its declared target.

The opened CIFAR-10-C diagnostic first asks whether each committed direction was correct and reports the corresponding action exposure. Tent makes two false ADAPT actions and one false FREEZE; EATA makes neither; SAR makes no FREEZE decisions. The secondary utility result is candidate-specific: Tent and EATA have lower point regret than either fixed policy, whereas SAR favors always-adapt. CCT-20 instead preserves frozen-model utility in all 45 cells with no ADAPT exposure. This is conservative frozen-model retention, not selective routing on an unseen natural shift. The appendices retain adverse, incomplete, and withheld studies so that this boundary does not depend on removing unfavorable evidence.

Newer painting-development evaluations retain the same limitation: the five-checkpoint cohort and separate reference-source candidates yielded universal abstention, with adverse queue and normalization-state diagnostics preserved as development evidence.

Sections 3–5 define the population target, derive the strict-commitment frontier, and establish the evidence-fibre audit floor. Section 6 introduces the separate empirical gate. Sections 7–8 report the controlled CIFAR diagnostic and CCT-20 retention result. Section 9 states the assumptions, missing evidence, and deployment limits.

KGA empirical decision flow

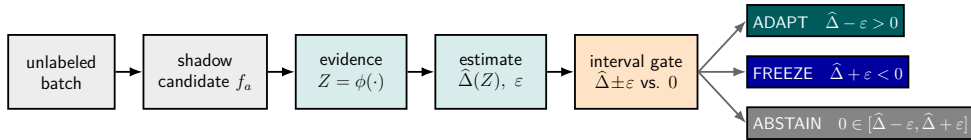


Figure 1: KGA’s empirical decision flow for one evaluation cell. A shadow adapter creates f_a from frozen f_0 . Labeled development cells fit the benefit predictor, and disjoint residual-calibration cells determine its error radius. The scored cell contributes label-free evidence only. An interval above zero selects ADAPT, one below zero selects FREEZE, and an interval crossing zero selects ABSTAIN. The figure defines the action rule; it does not establish interval coverage.

2 Related Work

2.1 Candidate Adaptation and Update Selection

Tent, EATA, SAR, CoTTA, MEMO, SHOT, and test-time training construct candidates through entropy objectives, filtering, regularization, temporal stabilization, augmentation, or auxiliary tasks (Wang et al., 2021; Niu et al., 2022, 2023; Wang et al., 2022; Zhang et al., 2022; Liang et al., 2020; Sun et al., 2020). Other methods decide where or when to update. TTSA selects modality-specific adapters (Chen et al., 2025). GALA selects layers and filters samples by gradient alignment (Sahoo et al., 2025). Informed Adaptation learns topological features associated with useful updates (Mutlu et al., 2026). KGA starts later in this pipeline: it compares one realized candidate with its frozen baseline. Learned routing and rollback are established ideas, not our novelty claim.

2.2 Label-Free Performance and Strategy Estimation

ATC, agreement-on-the-line, AETTA, and prediction-matrix statistics estimate performance from unlabeled outputs and suitable development information (Garg et al., 2022; Baek et al., 2022; Miller et al., 2021; Lee et al., 2024; Deng et al., 2023; Xie et al., 2024). The closest empirical comparator is Kim et al. (2024), whose TTA/ALine method uses agreement-based estimators after adaptation, supports hyperparameter selection, and compares adaptation strategies

without labeled OOD data.

KGA directly fits the paired score difference for one frozen/candidate pair and converts its prediction interval into a three-way action. Separate adapted- and frozen-model accuracy estimates can also be differenced; a synchronized current-policy comparison with official TTA/ALine and AETTA implementations remains missing.

2.3 Monitoring and Adaptation Controllers

POEM protects adaptation through entropy matching and online betting (Bar et al., 2024). Schirmer et al. (2025) monitor an adapting model’s running risk against a source-risk reference under a proxy-informativeness condition; a supervised precursor uses a labeled test stream (Podkopaev and Ramdas, 2022). Drift-to-Action Controllers use delayed labels and sequential evidence to gate adaptation, abstention, rollback, and retraining (Lamaakal et al., 2026). These methods act during an evolving stream. KGA instead decides whether one proposed candidate should be committed before it replaces the frozen state; it makes no anytime-valid risk claim.

2.4 Partial Identification and Robust Decisions

An identified set contains the values compatible with the observations and assumptions. Decision theory under partial identification studies actions when that set leaves payoffs ambigu-

ous. Christensen et al. (2026), for example, combine worst-case payoff uncertainty with statistical learning of identified parameters. K-Bound asks the narrower question of whether one strict benefit sign is uniform over the set. It neither solves the general optimal-decision problem nor claims interval thresholding as a new idea.

Indistinguishable problems are already central to domain-adaptation impossibility (Ben-David et al., 2010a,b). Our contribution is a disagreement-conditional label-kernel construction, the resulting clipped benefit set, and its residual-audit floor. The proof does not introduce label-free impossibility. Risk-estimation assumptions based on conditional independence (Steinhardt and Liang, 2016), label shift (Lipton et al., 2018; Garg et al., 2020), or disagreement discrepancy (Rosenfeld and Garg, 2023) restrict the compatible laws. Likewise, MMD-credal risk intervals impose RKHS and covariate-shift structure (Ariq, 2026). Each restriction can shrink the identified set. A converse theorem for a richer class does not contradict those results.

2.5 Marginal and Conditional Coverage

Selective classification and reject-option learning abstain on individual predictions (Geifman and El-Yaniv, 2017; Chow, 1970; Cortes et al., 2016). KGA instead abstains on an update. Its coverage-to-action proposition uses standard interval logic (Tibshirani et al., 2019; Barber et al., 2023; Bates et al., 2021; Angelopoulos et al., 2025). The statistical burden is to justify coverage for the intended unit, especially after selection or under shift.

Jin and Ren (2025)’s JOMI procedure targets coverage conditional on selection, assuming exchangeability and a compatible selection rule. Gibbs et al. (2025) obtain conditional guarantees over specified shift classes through a different calibration construction. KGA’s ordinary residual interval implements neither method. Its marginal false-adapt implication must therefore remain separate from error conditional on ADAPT and from coverage within each target environment. Learn Then Test addresses the

separate problem of configuration selection and multiplicity (Angelopoulos et al., 2025).

3 Setup, Notation, and Claim Levels

This section fixes the probability target, evaluation unit, model pair, and gate before defining actions or error events. Table 2 separates the population and empirical symbols used throughout.

3.1 Model Pair, Evidence, and Evaluation Unit

Let $(\Omega, \mathcal{F}, P_T)$ support a target draw (X, Y) , and let P_S denote the source law. For loss ℓ , define

$$R_T(f) = \mathbb{E}_{P_T}[\ell(f(X), Y)].$$

We fix a realized pair of measurable predictors for the population comparison. The frozen predictor is f_0 ; an adapter $\mathcal{A}$ uses an unlabeled batch $X_{1:m}$ to construct the fixed candidate $f_a = \mathcal{A}(f_0, X_{1:m})$. Population risk then integrates a fresh target draw while holding this pair fixed.

The gate may read label-free evidence

$$Z = \phi(X_{1:m}, f_0, f_a),$$

such as predictions, logits, and update statistics. It may not read the outcome of the cell it is about to score.

Scope of the matched-world observation experiment. The population evidence-law and randomized-decision statements use the same fixed observation experiment in each compatible world and the same fixed decision kernel. The observation law factors through the fixed input marginal; finite iid input sampling is one proved realization. Other dependent sampling designs require their own common-law premise. Randomized decisions require a common joint law of observation and random seed; an independent common seed is one sufficient construction, whereas equal seed marginals alone do not establish that joint law. Equality of one-input

Table 1: Neighboring approaches differ in inferential target, decision timing, and guarantee. The comparison is conceptual; the archived empirical ports are not an official synchronized benchmark.

Approach	Target	Decision timing	Guarantee relevant here
TTA/ALine; AETTA POEM; risk monitors	adapted-model performance running adaptation risk	after candidate evaluation during or after adaptation	performance estimate sequential monitoring under named assumptions
Partial-identification decisions	payoff set across compatible laws	decision under ambiguity	robust-decision context
K-Bound / KGA	population benefit set / pre- dicted cell benefit	before candidate commit- ment	strict frontier / coverage- conditional action

Table 2: Population and empirical quantities. β bounds a population residual by assumption; ε is an empirical prediction-error radius. They are not interchangeable.

Quantity	Level	Meaning
f_0, f_a	both	frozen predictor and realized shadow candidate
Δ	population	expected risk reduction $R_T(f_0) - R_T(f_a)$
$D, \mu_T(D)$	population	disagreement region and its target probability
M	population	label-free mean correctness-score margin on D
γ, β	population	hidden calibration residual and its externally declared absolute bound
Δ^{cell}	empirical	observed score difference on one predeclared evaluation unit
$\hat{\Delta}, \varepsilon$	empirical	predicted cell benefit and residual-based interval radius
FA_u	empirical	probability of ADAPT and a nonpositive cell benefit
FA_c	empirical	probability of nonpositive cell benefit conditional on ADAPT
FF_u	empirical	probability of FREEZE and a nonnegative cell benefit
FF_c	empirical	probability of nonnegative cell benefit conditional on FREEZE
Commitment rate	empirical	fraction of ADAPT or FREEZE decisions; not interval coverage

marginals does not imply equality for arbitrary world-dependent batch couplings. These are the formal model’s premises, not conclusions about the reported data. Holding the model pair fixed in this experiment does not justify conditioning on a pair adapted on that same batch. We do not establish that the transductive benchmark satisfies this observation model: outcome blindness alone supplies neither that conditioning argument nor iid evaluation.

An *evaluation cell* j is a predeclared collection of records $\mathcal{E}_j$ on which both models are scored. For a fixed higher-is-better score S , its measured benefit is

$$\Delta_j^{\text{cell}} = S(f_a; \mathcal{E}_j) - S(f_0; \mathcal{E}_j).$$

This is not the population benefit Δ . For accuracy it is an average of paired correctness differences. For macro-F1 it is a difference between two cell-level scores, not an average of

per-example losses. We therefore report metric scales separately.

3.2 Actions and Their Operational Meaning

Definition 1 (Decision semantics). The gate g maps its permitted evidence and frozen calibration artifacts to $\{\text{ADAPT}, \text{FREEZE}, \text{ABSTAIN}\}$. ADAPT commits f_a when an interval supports positive benefit. FREEZE rejects f_a when an interval supports negative benefit. ABSTAIN retains f_0 when neither strict direction is supported or the assessment is unavailable.

Freeze is a supported rejection; abstention is an unresolved assessment. An operator may attach monitoring, label acquisition, or review to an abstention. The experiments here evaluate only the frozen fallback and uncertainty record, not those additional interventions. If evidence

is missing, retaining f_0 must be logged as ABSTAIN when the interface returns an unresolved assessment. An integrity failure may instead raise an exception; the caller must preserve the frozen service and record the failure, not a certified FREEZE. Algorithm 1 describes the decision semantics after this integrity boundary has been handled.

The observed-cell oracle chooses the model with the larger score. Regret is the policy’s score shortfall from that oracle, averaged over the named cells with the stated weights. Always-freeze regret is $\max(\Delta_j^{\text{cell}}, 0)$; always-adapt regret is $\max(-\Delta_j^{\text{cell}}, 0)$. KGA serves the candidate only after ADAPT. Lower regret is better. A zero-benefit tie has zero regret for either model but supports neither strict sign.

3.3 Commitment, Interval Coverage, and Error

Let J be a random evaluation unit drawn from a predeclared deployment-unit distribution Q . The common probability space includes fitting and calibration randomness, candidate construction, J , and the records in $\mathcal{E}_J$. Write $\Delta^{\text{cell}} = \Delta_J^{\text{cell}}$ and $g = g_J$. The probabilities below are prospective marginal quantities under that joint law. When we instead summarize a finite recorded grid, we use the equal-weight empirical measure over its cells and call the result descriptive. It does not estimate the prospective law without an additional sampling design.

With this law and the gate fixed, we distinguish three measurements:

$$\text{Commit} = \mathbb{P}(g \in \{\text{ADAPT}, \text{FREEZE}\}), \quad (2)$$

$$\text{Cov}_{\text{cell}} = \mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon), \quad (3)$$

$$\text{FA}_{\text{u}} = \mathbb{P}(g = \text{ADAPT}, \Delta^{\text{cell}} \leq 0). \quad (4)$$

Conditional false adaptation is $\text{FA}_{\text{c}} = \mathbb{P}(\Delta^{\text{cell}} \leq 0 \mid g = \text{ADAPT})$, defined only when $\mathbb{P}(g = \text{ADAPT}) > 0$. The operational proposition below bounds only the marginal error, and only when the stated interval coverage premise holds. It does not bound FA_{c} . A gate that never adapts has zero false adaptations mechanically, so every error report includes action exposure. A false

FREEZE is defined analogously by $g = \text{FREEZE}$ and $\Delta^{\text{cell}} \geq 0$. Its conditional rate is undefined when no FREEZE action occurs.

The population theorem uses binary labels and 0/1 loss. The empirical gate can target a predeclared scalar score difference, including a multiclass metric, but that interface does not extend the binary theorem. The companion reviewer guide records the multiclass identity.

Six-symbol guide. M is the observable population-side disagreement margin; γ is the hidden target calibration residual; and β is an externally declared bound on $|\gamma|$. These quantities bracket population benefit Δ . At the empirical level, Δ^{cell} is the measured cell benefit, $\hat{\Delta}$ is its prediction, and ε is the calibrated prediction-error radius. Thus (M, γ, β) supports population reasoning, whereas $(\hat{\Delta}, \varepsilon)$ brackets Δ^{cell} ; the two uncertainty objects are not interchangeable.

4 Population Partial-Identification Frontier

The decision depends on one unknown sign: the sign of Δ . Target labels hide that sign. M supplies the visible direction, γ records what labels still hide, and β limits how large that hidden term may be. Together they yield the exact three-action frontier.

4.1 Disagreement-Region Model

The two predictors can differ in risk only where they disagree. We therefore isolate that region before introducing the score and its hidden residual.

Assumption 1 (Deployment setup). Let $Y \in \{0, 1\}$ and use 0/1 loss. Fix measurable binary predictors f_0 and f_a . Their disagreement region is $D = \{x : f_0(x) \neq f_a(x)\}$; assume $\mu_T(D) > 0$.

Let $s : \mathcal{X} \rightarrow [0, 1]$ be a fixed measurable candidate-correctness score fitted from source information. The target law has the measurable correctness kernel $\eta_a(x) = \mathbb{P}_T(f_a(X) = Y \mid X = x)$. Conditional on D , define

$$\begin{aligned} M &= \mathbb{E}[s(X) \mid D] - \frac{1}{2}, \\ \gamma &= \mathbb{E}[\eta_a(X) - s(X) \mid D]. \end{aligned} \quad (5)$$

Table 3: Validity obligations for the paper’s three claim levels. Each row states its target, required premise, supported conclusion, and excluded inference.

Claim level	Required premise	Supported conclusion	Not implied
Population frontier	binary 0/1 loss, fixed augmented evidence, declared residual class; richness for necessity	exact strict-sign frontier within that class	learning β from unlabeled data
Operational proposition	marginal interval coverage for a named scalar target and unit	marginal false commitment for that target	conditional, group, or any-time control
Empirical evaluation	fixed candidates, splits, metric, and action rule	measured cell regret, inclusion, and action exposure	population transfer or prospective confirmation

The class radius $\beta \geq 0$ is supplied externally. It defines $\mathcal{C}_\beta = \{P_T : |\gamma(P_T)| \leq \beta\}$. Because $s \in [0, 1]$, the feasible margin is $M \in [-\frac{1}{2}, \frac{1}{2}]$. The identities below do not assume that s is source-calibrated.

Remark 1 (Residual identity). By definition, $M + \gamma = \mathbb{E}[\eta_a(X) \mid D] - \frac{1}{2}$. This is an algebraic decomposition, not evidence that calibration transfers. We call γ the target disagreement-conditional calibration residual; conditioning can make it nonzero even without distribution shift. The companion reviewer guide gives the full counterexample and calibration qualification.

Definition 2 (Risk alignment). Evidence is risk-aligned when evidence-identical laws cannot have opposite nonzero benefit signs. This is weaker than uniform strict-sign identification: a fibre may still contain zero and positive benefit. We use the augmented evidence $\tilde{Z} = (Z, M)$.

Definition 3 (Strict directional soundness). At a fixed augmented-evidence law, ADAPT is sound when $\Delta(P) > 0$ for every compatible $P \in \mathcal{C}_\beta$, and FREEZE is sound when $\Delta(P) < 0$ throughout the fibre. A pointwise maximal rule commits exactly when one strict action is sound and abstains otherwise; $\Delta = 0$ supports neither strict claim.

4.2 Label-Kernel Freedom on the Disagreement Region

The central obstacle is now visible. Labels can change on D while the input distribution and all model outputs stay fixed. The next lemma isolates that freedom.

Lemma 1 (Label-kernel freedom on disagreement). *Fix $(\mu_T, P_S, f_0, f_a, s)$, and therefore fix D and M . Also fix the conditional label law outside D . For every measurable $\eta : D \rightarrow [0, 1]$, there is a target law with input marginal μ_T and $\mathbb{P}(f_a(X) = Y \mid X = x) = \eta(x)$ on D . Across these laws, every label-free observable formed from the input batch, fixed models, and model outputs under the declared common observation experiment has the same distribution. Meanwhile,*

$$\gamma = \mathbb{E}[\eta(X) \mid D] - \frac{1}{2} - M.$$

Proof. On D , one binary predictor outputs 0 and the other outputs 1. Assign conditional probability $\eta(x)$ to $Y = f_a(x)$ and $1 - \eta(x)$ to $Y = f_0(x)$. Retain the fixed label kernel outside D . This construction changes neither the input marginal nor any model output. The declared observation law factors through that fixed marginal, so its label-free evidence laws are preserved. Equation (5) gives the displayed expression for γ . $\square$

Consequently, every statistic formed from target inputs and fixed model outputs under this observation experiment has unchanged law, although the models’ relative correctness can change.

4.3 Benefit Reduction and Matched-Evidence Impossibility

The first result connects the decision target to the three population symbols. Because the models agree off D and binary correctness is complementary on D , benefit depends only on $M + \gamma$.

Lemma 2 (Disagreement-region reduction). *Under Assumption 1, with $\bar{a} := \mathbb{P}_T(f_a=Y \mid D)$,*

$$\begin{aligned}\Delta &= 2\mu_T(D)(\bar{a} - \tfrac{1}{2}), \\ \text{sign } \Delta &= \text{sign}(\bar{a} - \tfrac{1}{2}) = \text{sign}(M + \gamma).\end{aligned}$$

The identity $\text{sign } \Delta = \text{sign}(M + \gamma)$ holds for every target law; it does not assume $|\gamma| \leq \beta$.

Proof. Outside D , the two predictors agree and incur the same 0/1 loss. On D , exactly one is correct. The adapted and frozen accuracies conditional on D are therefore $\bar{a}$ and $1-\bar{a}$. Hence $\Delta = 2\mu_T(D)(\bar{a} - \frac{1}{2})$. The definitions of M and γ give

$$\begin{aligned}M + \gamma &= \left(\mathbb{E}_{\mu_T|D}[s] - \tfrac{1}{2}\right) + \mathbb{E}_{\mu_T|D}[\eta_a - s] \\ &= \mathbb{E}_{\mu_T|D}[\eta_a] - \tfrac{1}{2} = \bar{a} - \tfrac{1}{2}.\end{aligned}$$

Because $\mu_T(D) > 0$, the three quantities have the same sign. $\square$

The reduction tells us which hidden quantity controls the sign. The harder question is identification. Can two target worlds agree on every permitted observation while assigning opposite signs to Δ ? Inside the declared band, the answer is yes.

Theorem 1 (Interior matched-evidence impossibility). *Fix $\beta > 0$. For any feasible margin $M \in [-\frac{1}{2}, \frac{1}{2}]$ with $|M| < \beta$, there exist $P_T^+, P_T^- \in \mathcal{C}_\beta$ with the following properties:*

$$\begin{aligned}\text{Law}(\tilde{Z} \mid P_T^+) &= \text{Law}(\tilde{Z} \mid P_T^-), \\ M(P_T^+) &= M(P_T^-) = M, \\ \text{sign } \Delta_+ &= -\text{sign } \Delta_- \neq 0,\end{aligned}$$

where $\tilde{Z} = (Z, M)$. If $\beta = 0$, then $\mathcal{C}_0$ forces $\gamma = 0$ and $\text{sign } \Delta = \text{sign } M$; matched-evidence ambiguity therefore requires $\beta > 0$.

Proof. Choose $0 < \delta < \min\{\beta - |M|, \frac{1}{2}\}$. For $\theta \in \{+1, -1\}$, apply Lemma 1 with the constant correctness kernel

$$\eta_\theta(x) = \tfrac{1}{2} + \theta\delta, \quad x \in D,$$

and use the same fixed label kernel outside D . Denote the resulting target law by P_T^θ . The

input marginal, predictors, score, and source law are fixed, so M is the same under P_T^+ and P_T^- . Lemma 1 also gives the same law of Z in the two worlds. Consequently, $\text{Law}(\tilde{Z} \mid P_T^+) = \text{Law}(\tilde{Z} \mid P_T^-)$ for $\tilde{Z} = (Z, M)$.

It remains to verify class membership and the benefit signs. Under P_T^θ , $\bar{a}_\theta = \frac{1}{2} + \theta\delta$ and

$$\gamma_\theta = \theta\delta - M, \quad |\gamma_\theta| \leq \delta + |M| < \beta.$$

Thus both laws belong to $\mathcal{C}_\beta$. Lemma 2 yields $\Delta_\theta = 2\mu_T(D)\theta\delta$. The two benefits are nonzero and have opposite signs. $\square$

The evidence laws coincide exactly—their total-variation distance is zero—so this is exact non-identifiability, not a rate-dependent testing bound. Randomization cannot resolve two worlds that induce exactly the same evidence law. Controlling both directional errors therefore requires abstention.

Corollary 1 (Matched-evidence abstention lower bound). *Fix $\alpha \in (0, \frac{1}{2})$ and the shared evidence law from Theorem 1. Let g be any possibly randomized label-free rule. If, in every admissible world,*

$$\begin{aligned}\mathbb{P}[g = \text{ADAPT}, \Delta \leq 0] &\leq \alpha, \\ \mathbb{P}[g = \text{FREEZE}, \Delta \geq 0] &\leq \alpha,\end{aligned}$$

then $\mathbb{P}[g = \text{ABSTAIN}] \geq 1 - 2\alpha$.

Proof. The rule has the same action probabilities in both worlds. Its ADAPT probability is at most α in the negative world. Its FREEZE probability is at most α in the positive world. The three action probabilities sum to one. $\square$

4.4 Boundary Case

Inside the band, both strict signs may be feasible. At the boundary, zero benefit is feasible. These constructions differ, but both prevent a uniform strict direction.

Proposition 1 (Boundary case and closed-band abstention). *Fix $\alpha \in (0, \frac{1}{2})$. Under Definition 3, abstention is the pointwise maximal sound action throughout $|M| \leq \beta$. Any label-free rule that*

controls both marginal directional errors at level α must abstain there with probability at least $1 - 2\alpha$. Opposite nonzero benefit signs are asserted only when $|M| < \beta$.

Proof. For $|M| < \beta$, apply Theorem 1 and Corollary 1.

Now suppose $|M| = \beta > 0$. The declared class contains a zero-benefit world with $\gamma = -M$ and a strict-benefit world with $\gamma = 0$. These worlds have the same augmented-evidence law. In the zero-benefit world, both directional error events occur whenever the rule commits. Dual error control therefore gives $\mathbb{P}[\text{ADAPT}] \leq \alpha$ and $\mathbb{P}[\text{FREEZE}] \leq \alpha$. It follows that $\mathbb{P}[\text{ABSTAIN}] \geq 1 - 2\alpha$, and neither strict action is uniformly sound.

Finally, if $M = \beta = 0$, then $\mathcal{C}_0$ forces $\gamma = 0$ and $\Delta \equiv 0$. The same conclusion follows directly. $\square$

This convention is stricter than task regret. When $\Delta = 0$, adapt and freeze are risk-equivalent, but neither certifies a strict sign.

4.5 Exact Strict-Commitment Frontier

We can now answer the population question completely. The theorem separates four jobs: reduction, sufficiency outside the band, necessity inside the band, and maximality of the resulting rule.

Theorem 2 (Exact strict-commitment frontier). *Under Assumption 1, fix $(\mu_T, P_S, f_0, f_a, s)$ and let $\beta \geq 0$. Necessity and maximality hold for $\mathcal{C}_\beta$, or for a subclass closed under the constructions in Theorem 1 and Proposition 1; otherwise only sufficiency follows without an additional richness assumption.*

- (i) **Reduction.** For every P_T , $\text{sign } \Delta = \text{sign}(M + \gamma)$ (Lemma 2).
- (ii) **Sufficiency.** If $|M| > \beta$ and $P_T \in \mathcal{C}_\beta$, then $\text{sign } \Delta = \text{sign } M$.
- (iii) **Necessity.** If $|M| < \beta$, two laws in $\mathcal{C}_\beta$ have the same augmented-evidence law and opposite nonzero benefit signs. If $|M| = \beta > 0$, the admissible residual $\gamma = -\text{sign}(M)\beta$ gives

$\Delta = 0$. If $M = \beta = 0$, then $\Delta \equiv 0$ throughout $\mathcal{C}_0$. Thus $|M| \leq \beta$ supports no uniform strict direction.

- (iv) **Strict-commitment iff.** At the fixed augmented-evidence law, a strict action is uniformly directionally sound if and only if $|M| > \beta$. On the closed band, abstention is the pointwise maximal sound action in the sense of Definition 3.

The population rule $M > \beta \Rightarrow \text{ADAPT}$, $M < -\beta \Rightarrow \text{FREEZE}$, $|M| \leq \beta \Rightarrow \text{ABSTAIN}$ is the pointwise maximal sound rule on $\mathcal{C}_\beta$.

Proof. Part (i) is Lemma 2. For (ii), suppose $M > \beta$ and $|\gamma| \leq \beta$. Then $M + \gamma > 0$, so Lemma 2 gives $\Delta > 0$. The negative case is symmetric.

For (iii), Theorem 1 supplies opposite nonzero signs when $|M| < \beta$. At $|M| = \beta > 0$, the boundary law with $\gamma = -\text{sign}(M)\beta$ has zero benefit. When $M = \beta = 0$, every law in $\mathcal{C}_0$ has zero benefit. These cases preclude a uniform strict direction throughout the closed band.

Part (iv) combines (ii), (iii), and Proposition 1. For a nonempty compatible subclass, sufficiency gives the unique sound strict direction outside the band. This individual directional uniqueness requires a realized member: on an empty subclass, both universal directional statements are vacuous. The full compatible class is nonempty. The impossibility and boundary results rule out both strict directions on the band. $\square$

Outside the band, the residual cannot reverse M ; on the band, the same evidence admits a failed strict claim, so abstention is part of the identified answer. Let $d = \mu_T(D) > 0$. The full class identifies

$$\Delta(\mathcal{C}_\beta) = 2d[\max\{-\tfrac{1}{2}, M - \beta\}, \min\{\tfrac{1}{2}, M + \beta\}].$$

The bracket equals $[-\frac{1}{2}, \frac{1}{2}] \cap [M - \beta, M + \beta]$, and constant correctness kernels attain every point. A strict direction exists exactly when this interval excludes zero.

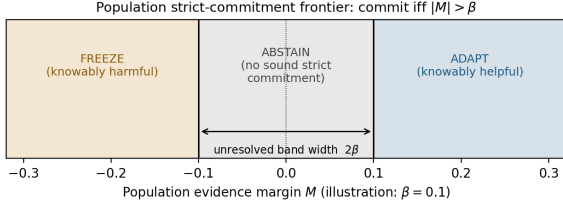


Figure 2: Population strict-commitment frontier for a fixed binary model pair under 0/1 loss. ADAPT is uniformly supported when $M > \beta$, FREEZE when $M < -\beta$, and neither strict direction is supported on $|M| \leq \beta$. This population result is not KGA’s numerical cell-level rule.

Region	Meaning	Action
$M > \beta$	strictly positive benefit	ADAPT
$M < -\beta$	strictly negative benefit	FREEZE
$ M < \beta$	both strict signs are feasible	ABSTAIN
$ M = \beta > 0$	zero benefit is feasible	ABSTAIN
$M = \beta = 0$	benefit is identically zero	ABSTAIN

4.6 Interpretation and Limits of the Frontier

The theorem is exact only for the declared class. Choosing $\beta = 0$ assumes a zero target disagreement-conditional residual. It is not a conservative default. Without a credible upper bound on $|\gamma|$, the frontier cannot support a robust strict action. More unlabeled samples do not close this gap because the missing information lies in the target label kernel.

Empirical abstention does not prove structural unknowability. A finite-sample interval may be wide because calibration data are scarce, prediction is poor, or transfer fails. This distinction leads to the next question: can the required population residual bound be learned from the same label-free evidence?

5 What Label-Free Evidence Cannot Learn

The population frontier becomes usable only after β has been declared. Can the same label-free evidence learn a smaller valid budget? An audit

must cover every residual value left unresolved on its evidence fibre.

5.1 Audit Setup and Evidence Fibres

Let W collect everything a deployment-time audit may read: the unlabeled target batch, fixed models, score function, and derived label-free statistics. It may also contain a source or calibration sample whose law does not depend on the target label kernel. A randomized procedure returns a measurable real-valued audit $\hat{\beta} = g(W, U) \geq 0$, where the seed U has the same distribution in every world and is independent of W in every world.

For a declared class $\mathcal{C}$ and an observable-law value z , define the evidence fibre

$$\mathcal{C}(z) = \{P \in \mathcal{C} : \text{Law}_P(W) = z\}$$

and its unresolved residual radius

$$\Gamma_z(\mathcal{C}) = \sup_{P \in \mathcal{C}(z)} |\gamma(P)|. \quad (6)$$

The fibre contains target laws that the audit cannot distinguish. Its radius is the smallest uniform bound on $|\gamma|$ that this observable law can support.

5.2 Exact Fibre-Wise Audit Floor

The answer is a floor, not an estimator. Any audit that is valid throughout one evidence fibre must cover the largest residual that remains possible in that fibre.

Theorem 3 (Exact label-free audit floor). *Fix a nonempty fibre $\mathcal{C}(z)$ with bounded attainable residuals $\{|\gamma(P)| : P \in \mathcal{C}(z)\}$ and $\delta \in [0, 1)$. Let $\mathbb{P}_z$ denote the common joint law of (W, U) on this fibre. Suppose the measurable nonnegative audit $\hat{\beta}$ satisfies $\mathbb{P}_P(|\gamma(P)| \leq \hat{\beta}) \geq 1 - \delta$ for every $P \in \mathcal{C}(z)$. Then*

$$\mathbb{P}_z(\hat{\beta} \geq \Gamma_z(\mathcal{C})) \geq 1 - \delta.$$

The constant audit $\hat{\beta} \equiv \Gamma_z(\mathcal{C})$ is valid with $\delta = 0$. Consequently, on the displayed high-probability event, an audited frontier can commit only where the frontier with radius $\Gamma_z(\mathcal{C})$ already commits.

Proof. The common joint law of (W, U) induces the same law for $\hat{\beta}$ in every world. Choose $P_n \in \mathcal{C}(z)$ so that $|\gamma(P_n)| \uparrow \Gamma_z(\mathcal{C})$. Under the common law, each event $\{\hat{\beta} \geq |\gamma(P_n)|\}$ has probability at least $1 - \delta$. These events decrease to $\{\hat{\beta} \geq \Gamma_z(\mathcal{C})\}$, so continuity from above proves the bound. The constant audit is valid by the definition of the supremum. Finally, $\hat{\beta} \geq \Gamma_z$ and $|M| > \hat{\beta}$ imply $|M| > \Gamma_z$. $\square$

The theorem does not prohibit useful budgets. An audit using the same observable law must cover every residual left possible on that fibre. A smaller bound requires labels or structure that genuinely refines the compatible set; $\Gamma_z(\mathcal{C})$ is an oracle fibre benchmark, not a general finite-batch estimator.

Declared budget and abstention radius. For the full declared class $\mathcal{C}_\beta = \{P : |\gamma(P)| \leq \beta\}$ with $0 \leq \beta \leq \frac{1}{2}$, feasible $M \in [-\frac{1}{2}, \frac{1}{2}]$, and positive disagreement mass, the label-kernel construction in Lemma 1 gives

$$\Gamma_z(\mathcal{C}_\beta) = \beta. \quad (7)$$

Thus the least valid fibre-wide residual bound equals the population abstention radius, β . This connection does not turn the declared β into an estimate.

When calibration labels do not shrink the target fibre. A labeled calibration domain does not reduce the target fibre unless the declared class links its label kernel to deployment. Lemma 1 can otherwise vary deployment labels while leaving the calibration sample unchanged. Labels therefore help through representativeness or structural coupling, not merely through their existence.

What additional structure can buy. Extra information can shrink the fibre, but it changes the problem. The available routes are representative deployment labels, exchangeable historical deployment episodes, a structural cross-domain coupling, or a domain-specific physical bound. Each gain comes from labels or assumptions, not the current unlabeled batch.

5.3 Implications for KGA

KGA does not estimate M , γ , or β , and does not apply $|M| > \beta$ numerically. It predicts Δ^{cell} and uses an empirical radius ε . Transferring cell-level coverage to population benefit requires an additional sampling-error argument that the reported experiments do not provide. The next section therefore changes the question from population identification to deployment-time prediction.

6 KGA: An Empirical Benefit-Interval Gate

For the current evaluation unit, can KGA support which side of zero the benefit lies on? It predicts cell benefit from label-free evidence and attaches a residual-calibrated radius.

6.1 Shared Decision Logic, Different Targets

Scope of KGA. KGA predicts measured evaluation-cell benefit Δ^{cell} , not population benefit Δ . Its radius ε describes empirical prediction error under a named calibration protocol. It neither estimates the population residual bound β nor establishes population-risk coverage.

Both layers commit only when their respective uncertainty set supports one strict sign, but they use different targets:

$$\begin{aligned} \Delta &= 2\mu_T(D)(M + \gamma), \\ \Delta^{\text{cell}} &= S(f_a; \mathcal{E}) - S(f_0; \mathcal{E}), \quad \hat{\Delta} = h_\theta(Z). \end{aligned} \quad (8)$$

Development cells fit h_θ , and disjoint residual-calibration cells supply residuals. A scored cell contributes Z , but not its outcome. The empirical coverage premise is

$$\mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon) \geq 1 - \alpha. \quad (9)$$

Proposition 2 applies only when this premise is justified.

6.2 The Coverage-to-Action Proposition

Proposition 2 (Coverage-to-action implication). *Fix the scalar benefit target and its evaluation unit before calibration. Let B and $\hat{B}$ be finite real random variables, and let $\varepsilon \in [0, \infty]$ be a measurable random radius on a common probability space containing all calibration, benefit-model fitting, candidate-construction, and evaluation-unit randomness. The prediction $\hat{B}$ and radius ε must be available without reading the new unit's evaluation labels. For $\alpha \in (0, 1)$, suppose the interval satisfies marginal coverage*

$$\mathbb{P}[|\hat{B} - B| \leq \varepsilon] \geq 1 - \alpha.$$

Define

$$g = \begin{cases} \text{ADAPT} & \hat{B} - \varepsilon > 0, \\ \text{FREEZE} & \hat{B} + \varepsilon < 0, \\ \text{ABSTAIN} & \text{otherwise.} \end{cases}$$

Then, marginally over that common probability space,

$$\begin{aligned} \mathbb{P}[g = \text{ADAPT}, B \leq 0] &\leq \alpha, \\ \mathbb{P}[g = \text{FREEZE}, B \geq 0] &\leq \alpha. \end{aligned}$$

Proof. On the event $\{|\hat{B} - B| \leq \varepsilon\}$, the condition $\hat{B} - \varepsilon > 0$ implies $B > 0$, so a false adapt requires the complement of this event, which has probability at most α . The freeze case is symmetric. $\square$

For the empirical panels, $B = \Delta^{\text{cell}}$ and $\hat{B} = \hat{\Delta}$. A population-risk application instead sets $B = \Delta$ and requires coverage for that quantity; cell-outcome coverage alone does not provide it.

Remark 2 (Scope of the implication). Proposition 2 controls marginal error for the declared target. It does not control error conditional on commitment, group-specific coverage, candidate-selection multiplicity, or repeated-time behavior. The coverage premise needs separate justification. When it is unsupported, a declared fallback may retain f_0 and record ABSTAIN, but that response does not prove f_0 safe or optimal and is

not a certified FREEZE decision. For this certificate, a population sign claim requires interval coverage for population benefit itself, or a justified cell-to-population transfer bound. Once that coverage premise holds, the event-containment argument applies directly; it does not need an additional risk-alignment assumption.

6.3 Cell-to-Population Transfer

Cell coverage alone does not imply population coverage. The following proposition states the additional sampling premise needed to change the target while preserving the three-way decision semantics. It is a conditional extension, not a replacement of KGA's cell-level calibration. The evaluation unit, target-environment population, model-pair construction protocol, score functional, and units must be declared before calibration.

Proposition 3 (Cell-to-population transfer). *Let $\hat{\Delta}$, Δ^{cell} , and Δ be real-valued measurable quantities on a common probability space, with cell and population benefit defined for the same model pair and score functional. Let $\varepsilon, b \geq 0$ be finite measurable radii. Suppose that $\alpha_{\text{cell}}, \delta \in [0, 1)$, $\alpha_{\text{cell}} + \delta < 1$, and*

$$\begin{aligned} \mathbb{P}(|\hat{\Delta} - \Delta^{\text{cell}}| \leq \varepsilon) &\geq 1 - \alpha_{\text{cell}}, \\ \mathbb{P}(|\Delta^{\text{cell}} - \Delta| \leq b) &\geq 1 - \delta. \end{aligned} \tag{10}$$

Then, without requiring independence of these two events,

$$\mathbb{P}(|\hat{\Delta} - \Delta| \leq \varepsilon + b) \geq 1 - \alpha_{\text{cell}} - \delta. \tag{11}$$

If $\hat{\Delta}$, ε , and b are available before the new evaluation labels, define $L_{\text{pop}} = \hat{\Delta} - (\varepsilon + b)$ and $U_{\text{pop}} = \hat{\Delta} + (\varepsilon + b)$. The rule

$$g_{\text{pop}} = \begin{cases} \text{ADAPT}, & L_{\text{pop}} > 0, \\ \text{FREEZE}, & U_{\text{pop}} < 0, \\ \text{ABSTAIN}, & \text{otherwise} \end{cases} \tag{12}$$

satisfies

$$\begin{aligned} \mathcal{E}_{\text{pop}} &= \{g_{\text{pop}} = \text{ADAPT}, \Delta \leq 0\} \\ &\quad \cup \{g_{\text{pop}} = \text{FREEZE}, \Delta \geq 0\}, \end{aligned} \tag{13}$$

$$\mathbb{P}(\mathcal{E}_{\text{pop}}) \leq \alpha_{\text{cell}} + \delta.$$

Proof. On the intersection of the two premise events, the triangle inequality gives $|\hat{\Delta} - \Delta| \leq |\hat{\Delta} - \Delta^{\text{cell}}| + |\Delta^{\text{cell}} - \Delta| \leq \varepsilon + b$. The complement of that intersection has probability at most $\alpha_{\text{cell}} + \delta$ by the union bound. On the intersection, $L_{\text{pop}} > 0$ implies $\Delta > 0$ and $U_{\text{pop}} < 0$ implies $\Delta < 0$, so either false strict direction requires failure of at least one premise event. $\square$

The bound is marginal over the declared evaluation unit and its protocol randomness, not conditional on commitment, groupwise, or anytime-valid. The radii need separately justified coverage premises; computing their sum does not establish those premises. No such sampling-error bound is established for the reported panels, whose gates continue to use ε , not $\varepsilon + b$. An unavailable premise or non-finite radius therefore supports no population certificate: the operational fallback is ABSTAIN while retaining f_0 , not a certified FREEZE. The sampling radius b is distinct from the population residual bound β . The companion population-transfer discussion records the sampling and score-functional restrictions.

6.4 Evidence, Calibration, and the Action Rule

Each candidate-specific model uses label-free summaries of confidence, class balance, model disagreement, distribution shift, and update magnitude. Natural tracks retain protocol-specific models and schemas. The companion evidence inventory records their exact specifications.

Given residuals $r_i = |\hat{\Delta}_i - \Delta_i^{\text{cell}}|$, ordered as $r_{(1)} \leq \dots \leq r_{(n)}$, the order-statistic rule is

$$k = \lceil (n+1)(1-\alpha) \rceil, \quad \varepsilon = \begin{cases} r_{(k)}, & k \leq n, \\ +\infty, & k > n. \end{cases} \quad (14)$$

The rank rule has split-conformal coverage when calibration and next-unit scores are jointly exchangeable under the full fitting and candidate-construction protocol (Tibshirani et al., 2019; Barber et al., 2023); here n counts cells, not images. Disjoint fit, calibration, and check labels

Algorithm 1 KGA decision for one proposed candidate

Require: Frozen f_0 , adapter $\mathcal{A}$, unlabeled X , frozen h_θ and calibration artifacts

- 1: Create $f_a \leftarrow \mathcal{A}(f_0, X)$ in isolated shadow state
 - 2: Extract $Z = \phi(X, f_0, f_a)$ and validate the assessment inputs
 - 3: **if** the assessment is unavailable **then**
 - 4: retain f_0 , log the reason, and return ABSTAIN
 - 5: **end if**
 - 6: Obtain ε from the calibrated residual rule in Eq. (14)
 - 7: $\hat{\Delta} \leftarrow h_\theta(Z)$
 - 8: $L \leftarrow \hat{\Delta} - \varepsilon$; $U \leftarrow \hat{\Delta} + \varepsilon$
 - 9: **if** $L > 0$ **then**
 - 10: commit f_a and return ADAPT
 - 11: **else if** $U < 0$ **then**
 - 12: discard f_a and return FREEZE
 - 13: **else**
 - 14: retain f_0 , log uncertainty, and return ABSTAIN
 - 15: **end if**
-

prevent the specified reuse; this does not establish exchangeability. The painting-development inclusion rates below are descriptive checks on opened cells, not validation of group-conditional, repeated-use, or population-risk coverage.

A deterministic five-fold cross-fit is keyed by stable cell IDs. For each score fold, its complement is divided into disjoint estimator-fit and residual-calibration subsets. Fit outcomes determine the predictor, and the exact-rank rule on calibration residuals determines the radius. No scored outcome can affect its own prediction, radius, or action; the whole-pipeline invariance test verifies this property. This separation blocks direct outcome flow but does not create independent environments or establish exchangeability. Natural tracks use separate partitions only where their archived protocols record them.

For radius 0.01, predictions 0.04, -0.04 , and 0.005 produce ADAPT, FREEZE, and ABSTAIN, respectively. These values illustrate the rule; they are not experimental results.

Runtime includes shadow adaptation, two-model inference, evidence extraction, and storage; comparable end-to-end latency measurements are unavailable.

7 Primary Experimental Protocols

7.1 Two Primary Questions and the Evidence Hierarchy

The CIFAR-10-C diagnostic asks whether KGA distinguishes helpful from harmful updates on an opened constructed grid. CCT-20 asks whether a frozen gate avoids measured candidate harm across camera locations. The first exercises directional branches retrospectively; the second evaluates governed retention and may pass without ADAPT exposure.

The opened, dependent, constructed CIFAR-10-C diagnostic (Hendrycks and Dietterich, 2019) uses six corruption families at severities 1 and 5. It crosses three batch sizes (200, 16, and 8), three compositions (IID, imbalanced, and single-class), two update schedules (10 steps at 10^{-3} and 50 at 2.5×10^{-3}), and two repeats: 432 cells per RNG seed and 2,160 per candidate over seeds 0–4.

Tent, EATA, and SAR are local ports applied to one ResNet-18 checkpoint; the five seeds repeat streams and adapter randomness, not checkpoint training. Point estimates weight cells equally; family sensitivity first averages within each family and then weights the six means equally.

The recorded candidates are local, protocol-matched composite operating points rather than untouched reproductions of published defaults. They combine evaluation-batch BatchNorm statistics with normalization-affine updates; Appendix D records the contract and unexecuted BatchNorm-only control.

The three-way cross-fit removes each scored fold’s outcomes from fitting and residual calibration, but does not make the shared checkpoint, corpus, or cells independent. This is a retrospective diagnostic, not prospective confirmation or

an untouched deployment trial.

7.2 CCT-20 Design and Locked Utility-Preservation Endpoint

The CCT-20 study (Beery et al., 2018) crosses 5 independent checkpoints with 9 target locations, giving 45 cells and 6950/16325 probe/evaluation images (23275 total). Set-membership top-1 accuracy credits any archived category, not an official single-label target. Official Tent makes one non-episodic Adam update per batch at 10^{-3} ; pre-update logits score it, state carries within a cell, and resets between cells.

Individual outcomes remained unopened until one-shot scoring, but aggregate target-label metadata had previously been inspected; CCT-20 was therefore cell-outcome-unopened and internally sealed, not globally label-unopened or publicly preregistered. Ridge fitting used 10 rows from two camera locations and calibration used 45 rows across nine others; transfer to target locations is assumed.

For fixed policy b , G_b is mean baseline regret minus mean KGA regret. The limited, predeclared utility check uses nominal pointwise 95% intervals and passes when their lower endpoints satisfy

$$L_{\text{adapt}} > 0, \quad L_{\text{freeze}} > -0.005. \quad (15)$$

The second threshold is frozen-policy noninferiority; this check can pass when KGA always serves f_0 and is not a population guarantee.

The separate strong-success rule requires complete data for all 45 cells, nominal 97.5% Bonferroni intervals above zero against both fixed policies, both exact nine-location tests surviving Holm at 0.05, ADAPT and FREEZE rates at least 0.10, and total commitment at least 0.30. It therefore requires interval-supported improvement over both fixed policies; any failure precludes strong success.

The CCT utility table uses pointwise 95% intervals; the separate companion audit uses 97.5% Bonferroni intervals. They are not alternate versions of the same interval.

Table 4: Evidence status and admissible claim. “Primary” means central to the paper, not prospectively confirmatory; unfavorable and incomplete studies remain in scope.

Evidence	Design and status	Question it can answer
CIFAR-10-C	opened, dependent, constructed; cell-outcome-disjoint cross-fit	retrospective candidate-specific routing
CCT-20	cell-outcome-unopened, internally sealed execution; aggregate target metadata previously inspected	conservative retention under the locked endpoint
So2Sat-LCZ42	development stop before target access	candidate failed eligibility
DomainNet painting	opened development; five checkpoints on shared cells and separate reference-source candidates; fixed follow-up diagnostics	candidate feasibility, abstention, and mechanism limits; not confirmation
Auxiliary replays	opened or one-sided diagnostics	transfer limits and supporting behavior
Invalid or partial records	PACS partial; iWildCam metric invalid; historical ports superseded; Office-Home serialization invalid	audit history only

Scoring and inference. Actions are fixed before outcomes are joined, but candidate TTA is transductive because it reads unlabeled evaluation inputs. CIFAR sensitivity resamples the six corruption-family means conditional on the archived checkpoint. CCT-20 uses a paired product bootstrap over checkpoint rows and location columns; its separate exact sign-flip reference calculation uses nine location means and requires a stronger joint sign-symmetry condition than exchangeability alone. All reported levels are nominal. The companion inference audit gives the metrics, sign convention, full assumptions, and retrospective status (Hemerik and Goeman, 2018).

8 Primary Results

8.1 CIFAR-10-C: Directional Actions and Candidate-Specific Utility

Table 5 first exposes the ADAPT/FREEZE/ABSTAIN record under one current rule. Tent makes two false ADAPT decisions and one false FREEZE; EATA makes neither; SAR makes no FREEZE decision, so that error direction is unexercised. These counts require their action and all-cell denominators below. Utility is a secondary measured consequence: Tent and EATA have lower point regret than both fixed policies, whereas always-adapt has lower regret than

KGA for SAR. The panel therefore supports only candidate-specific, retrospective action and utility results.

Tent regret is 0.0019, compared with 0.0080 for always-adapt and 0.1239 for always-freeze. EATA’s values are 0.0016, 0.0033, and 0.1313. Their ADAPT/FREEZE/ABSTAIN counts are 1,091/337/732 and 1,207/105/848, respectively. Tent makes two false ADAPT decisions; EATA makes none. SAR makes 1,414/0/746 decisions. Its KGA regret is 0.0018, compared with 0.0003 for always-adapt and 0.1405 for always-freeze. Abstention can prevent harmful updates while still forgoing helpful ones.

The nominal six-family intervals distinguish Tent from EATA. Gaps are baseline regret minus KGA regret, so positive values favor KGA. Tent’s gap versus always-adapt is 0.0061, with nominal family-bootstrap interval $[0.0021, 0.0095]$; its interval versus always-freeze is also strictly positive. EATA’s interval versus always-adapt crosses zero, $[-0.0013, 0.0044]$, although its interval versus always-freeze is positive. Tent therefore has the favorable nominal two-baseline pattern; EATA does not.

The retrospective reference tests do not make either pattern confirmatory. Across the six candidate-by-baseline contrasts in the retrospective comparison, Tent’s Holm values are 0.140625 against always-adapt and 0.09375 against always-freeze. EATA’s are 0.3125 and 0.09375, respectively. The replay, family unit, tests, and multi-

Table 5: Opened, dependent CIFAR-10-C diagnostic over 2,160 equally weighted cells per candidate. Lower regret is better. A/F/U gives action exposure; directional errors use the corresponding action denominator. Tent’s all-cell errors are 2/2,160 for false ADAPT and 1/2,160 for false FREEZE. All cells reuse one checkpoint, so this is retrospective rather than independent environment validation. Canonical panel SHA-256: 35d4c165843de1ece3cb35ffb4ac50dbfcbfa33b646754e2b6d133fbdfa78c6e.

Candidate	n	KGA	Adapt	Freeze	A/F/U	False A/A	False F/F	Commitment
Tent	2160	0.0019	0.0080	0.1239	1091/337/732	2/1091	1/337	0.6611
EATA	2160	0.0016	0.0033	0.1313	1207/105/848	0/1207	0/105	0.6074
SAR	2160	0.0018	0.0003	0.1405	1414/0/746	0/1414	–	0.6546

plicity analysis were selected after the grid was open, and all six families share one checkpoint. The companion family-sensitivity table reports every interval and reference value. Nominal interval exclusion and retrospective Holm adjustment answer different questions; neither establishes prospective coverage.

8.2 Retrospective Interval Diagnostics

Retrospective inclusion is 1,976/2,160 for Tent, 1,950/2,160 for EATA, and 1,949/2,160 for SAR. These values describe opened cross-fit cells; they are not independent validation of coverage on new environments. The companion inference appendix reports interval widths and family-level diagnostics. Tent’s two false ADAPT actions correspond to marginal error 2/2160 and conditional error 2/1091. It also makes one false FREEZE among 337 FREEZE decisions. EATA makes no false ADAPT and no false FREEZE. SAR makes no false ADAPT and no FREEZE decisions, so its conditional false-freeze rate is undefined. Exposure and denominators are essential: a zero count alone is not evidence of safety.

8.3 CCT-20: Conservative Retention, Not Selective Routing

All 45 checkpoint–location cells completed. KGA made 0/44/1 ADAPT/FREEZE/ABSTAIN decisions. All 45 served predictions used the frozen model, but the sole helpful cell received FREEZE, yielding one false FREEZE among 44

FREEZE actions (1/45 overall). Tent helped in 1 cell, tied in 0, and harmed in 44; 43 harmful cells received FREEZE, and the remaining harmful cell received ABSTAIN. The study therefore passes only the locked utility-preservation endpoint and does not demonstrate selective routing or strong success.

KGA matched always-freeze and avoided the measured degradation from always-adapt. The unchanged locked rule therefore returns pass on the stored cells (generated flag: yes). The archived verdict token remains SAFE UTILITY ONLY, but the supported description is utility preservation only. This is a scoped endpoint, not a finite-sample population-safety guarantee.

The complete strong-success audit is more informative. Completeness passed. Against always-adapt, the nominal 97.5% Bonferroni interval is $[0.08501071, 0.31066374]$ and the Holm-adjusted location sign-flip value is 0.00390625; both pass their locked thresholds. Against always-freeze, the contrast and interval are zero and the Holm value is 1; neither supports strict improvement. FREEZE exposure and total commitment exceed 0.10 and 0.30, but ADAPT exposure is zero and fails its 0.10 threshold. The strong-success criterion therefore fails. This is conservative frozen-model retention, not learned selective routing. The false-adapt count (0/45) is mechanically zero because ADAPT exposure is zero.

Every development cell was harmful for Tent, so the fit saw no helpful example. Only two camera locations supplied fitting data, and calibration-to-target transfer is unverified. The

Table 6: CCT-20 action record and locked utility-preservation endpoint over 45 checkpoint–location cells. Positive policy contrasts favor KGA. The 95% intervals belong to the utility endpoint; Appendix G reports the separate 97.5% Bonferroni strong-success audit.

Item	Result
Actions	0 ADAPT / 44 FREEZE / 1 ABSTAIN
Measured effects	1 helpful / 0 tied / 44 harmful
False FREEZE	1/44 conditional; 1/45 overall
Locked endpoint	Pass: utility preservation only
Strong success	Fail: no ADAPT exposure or improvement over always-freeze

Comparator	Mean contrast	Nominal 95% CI	Required lower bound
Always adapt	0.1819	[0.0937, 0.2914]	$L > 0.0000$
Always freeze	0.0000	[0.0000, 0.0000]	$L > -0.0050$

complete location ledger and primary inference appear in the companion CCT-20 ledger. We infer no selective-routing success from this result.

8.4 Development Stops and Adverse Diagnostics

So2Sat-LCZ42 (Zhu et al., 2020) screened Tent and SAR in a development-only, multi-city, multi-checkpoint feasibility study. Tent produced both helpful and harmful development cases, but neither candidate passed every pre-declared eligibility check. The study therefore stopped before gate calibration. No gate-calibration rows or target inputs, pixels, or labels were accessed, so there is no target natural-shift score. The gate-fit outcomes are now open; re-tuning them cannot create fresh confirmation.

DomainNet painting development. Five separately trained checkpoints were evaluated on the same 24 check cells per checkpoint. KGA abstained on all 120 seed-cell records; candidate effects were helpful in 12, harmful in 15, and tied in 93. Equal-seed regret was 0.110677 percentage points for both KGA and always-freeze, versus 0.123547 for always-adapt. These repetitions do not create 120 independent environments. Using a separate reference-source ResNet-50 checkpoint, a one-pass local entropy candidate restored the source BatchNorm running buffers before inference. Its 24-cell check panel had 7 helpful, 14 harmful, and 3 tied cells.

A separate AdaContrast-derived single-process recipe had 1 helpful and 23 harmful cells. Both gates abstained on all 24 cells. The two recipes produced different frozen prediction totals and are not a matched adapter comparison. These are opened-development negatives, not selective-routing or unseen-shift coverage evidence. Fixed queue and normalization-state diagnostics did not rescue the evaluated candidates; their scope and separate numerical summaries appear in the companion DomainNet development section.

The auxiliary studies expose further limits. ImageNet-R and PACS favor always-adapt, while ImageNet-C depends strongly on candidate and operating point. Office-Home and RxRx1 mainly retain the frozen model; the opened Camelyon17 OOD panel adapts throughout. The companion auxiliary-results section reports valid diagnostic rows on separate metric scales. It also explains why iWildCam, the historical POEM/AETTA ports, and the focused Office-Home route cannot support current headline comparisons.

ImageNet-C SAR operating point. The ImageNet-C SAR result uses a local port at learning rate 4×10^{-3} , which is $16\times$ the published 2.5×10^{-4} setting. The port adapts the recorded BatchNorm affine parameters, including the upper ResNet block that the official default excludes. It also combines target-batch BatchNorm-statistics replacement with gradient updates. This is an aggressive local diagnostic,

not an official paper-default configuration. We ran no matched control at SAR’s published operating point, so the result concerns this port and cannot be read as evidence against SAR as published.

ImageNet-R backbone pattern. KGA has higher regret than always-adapt on 8 of 10 backbones. On `efficientnet_b0`, KGA improves over always-adapt but ties always-freeze at zero regret. On `swin_t`, it also improves over always-adapt but ties always-freeze at 0.00473958 regret. Thus KGA has lower regret than both fixed policies on 0 of 10 backbones. Four of the eight adapt-favoring backbones have a 0% harmful-update base rate, so routing cannot improve on always-adapt there. This opened, descriptive pattern is consistent with routing being most useful when harmful updates occur. It is not one deployable multi-backbone policy or confidence-supported natural-shift evidence.

9 Limitations and Broader Impact

The measured action record does not demonstrate universal safety. The theory is exact relative to its declared class, whereas the empirical intervals are protocol-scoped and require separate coverage assumptions. The recorded experiments measure directional errors and utility only for the fixed candidates, metrics, and evaluation units studied. Neither layer establishes unconditional deployment safety.

9.1 Formal Assumptions and Empirical Calibration Scope

The population frontier is exact within its declared rich binary class. A credible β must come from domain knowledge, target labels, or a stated transfer assumption. A narrower class may permit stronger decisions. An unsupported bound cannot justify a strict population claim. The substantive object is the class-dependent feasible set, not the inequality by itself.

The development-based surrogate failed.

We tested estimating β from source-like development cells rather than declaring it. The estimate is $1.4\text{--}50\times$ too small on real evaluation cells (24–73% of cells fall outside the declared class $\mathcal{C}_\beta$; commit error 0.4–16.6%), and returns zero commitments on all 405 ImageNet-C cells in 5 of 10 configurations. We therefore withdraw the estimation route and declare β from domain knowledge instead.

This test used an operational development proxy; it could not observe the theorem’s target disagreement-conditional residual directly. The repository retains the reported result text but not the named raw beta-sweep artifact. The negative result is therefore historically documented but not presently regenerable from the canonical bundle. KGA neither estimates nor numerically uses β .

For KGA, empirical interval coverage needs separate support at the intended decision unit. A larger ε reduces commitment but may increase regret when f_0 is worse. A smaller ε increases exposure and the cost of calibration failure. Changing the adapter, checkpoint, feature schema, or update settings changes the prediction problem. Each change requires recalibration or a justified transfer argument.

Repeated deployment needs a separate guarantee.

A predeclared policy does not by itself create exchangeability or time-uniform safety. Reusing calibration data, replacing the frozen baseline after an accepted update, or selecting later candidates from earlier outcomes changes the joint process. Adaptive conformal inference studies online coverage frequency with feedback (Gibbs and Candes, 2021); it does not automatically give time-uniform control of KGA’s false commitments. Selection-conditional and group-conditional guarantees require additional methods, as Section 2 explains.

9.2 Evidence Needed for Prospective Confirmation

The controlled results reuse one checkpoint, one corpus, and related corruption cells. Their cell-

outcome-disjoint split fixes a leakage defect; it does not remove dependence. The interval inclusion table is retrospective, not an independent calibration test. The natural/development record remains retention-heavy or adverse: CCT-20 has no ADAPT exposure, So2Sat stops during development, and the painting-development gates abstain throughout. Five-checkpoint repetition documents development behavior but supplies neither independent unseen environments nor a prospective routing success. Empirical abstention may reflect weak prediction, limited data, a wide ε , or transfer failure. It does not establish population non-identifiability.

The search history is substantial. One superseded census reported 1,387 determinations and 326 positives (23.5%). A broader archived scan reported 1,427 determinations and 345 positives (24.2%). Neither census froze a reproducible family definition, file list, and repository state. They are overlapping historical search summaries, not two predeclared families and not the multiplicity family for the current panel. The controlled results remain explicitly post hoc.

Table 7 states the four missing tests and the claim each would inform. They cannot be supplied by reclassifying the existing panel.

The companion prospective-confirmation protocol specifies a future protocol.

9.3 Operational Risks and Deployment Scope

Rejecting an update can prevent adaptation harm. A frozen fallback can prevent adaptation-induced harm while still preserving a model that is inaccurate, biased, or unsuitable for the application. Treating abstention as a safety certificate would be especially risky. A deployment should retain the assessment reason and route unresolved cases to domain-appropriate review or evidence acquisition. Our experiments do not validate clinical, safety-critical, or autonomous operation. We claim neither a held-out physical camera deployment nor a common end-to-end latency result.

The maintained software distinguishes an unavailable assessment from a supported nega-

tive interval. Its tests and proof artifacts document a scoped validity, commitment-control, and rollback layer. They do not certify every data pipeline or production system. The appendices record reproduction instructions, formal scope, excluded historical extensions, and release checks.

10 Conclusion

K-Bound separates missing target-label information from estimable uncertainty. For a fixed binary model pair under 0/1 loss, a declared residual class induces a feasible set of population benefits. That set supports a strict direction exactly outside the closed band $|M| \leq \beta$. The evidence-fibre audit theorem explains why additional functions of the same label-free evidence cannot remove residual ambiguity that remains on the fibre.

KGA addresses a different target. It predicts measured evaluation-cell benefit and uses a separately calibrated interval to ADAPT, FREEZE, or ABSTAIN. Its empirical criterion is directional action correctness, with exposure and false-action denominators reported separately. The utility preservation endpoint is reported separately, not as proof of universal safety. The current CIFAR-10-C diagnostic exercises both directional actions retrospectively, whereas CCT-20 demonstrates conservative all-frozen retention rather than selective routing. These results support evidence-aware commitment under recorded protocols, not unconditional safety or population transfer. The newer opened development record likewise shows abstention and unrecovered candidate harm.

The practical lesson is simple: commit a proposed update only when the relevant target, evidence, and assumptions support a direction; otherwise preserve the fallback and record the unresolved uncertainty.

Table 7: Evidence needed for prospective confirmation. Each row names a missing test rather than reinterpreting a completed diagnostic.

Missing test	Question it would answer	Present boundary
Synchronized official comparators	Does the current policy add value beyond official TTA/ALine, AETTA, POEM, and simple benefit gates?	Only historical local-port diagnostics exist.
New natural shift with helpful and harmful candidates	Can a frozen policy exercise nonzero ADAPT and FREEZE exposure prospectively?	Current natural tracks are one-sided, retention-heavy, or stopped.
Untouched-group coverage evaluation	Do interval inclusion and directional errors transfer beyond opened cross-fit cells?	No held-out group-coverage result is available.
Independent checkpoints and environments	Does the inference extend beyond one checkpoint and related constructed conditions?	Five separately trained checkpoints have development results on shared painting cells; independent-environment prospective inference remains unavailable. The primary CIFAR grid reuses one checkpoint and corpus.

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